Grid Electrification and Energy Access in India

A Risk-Based Perspective on Energy Transitions in the Anthropocene

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MSc in Quantitative Finance (Data Science Track)

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20/3/2026

A report submitted as part of the requirements of SSOC600 Energy Transitions for Sustainability individual research report

Abstract

India has achieved near-universal electrification. The current challenge is to ensure that electricity is reliable, affordable, and sustainable, especially amid rising demand, volatile fossil-fuel markets, geopolitical risks, mineral supply constraints, and climate shocks. This paper conceptualises grid electrification and energy access as a stochastic risk-management and portfolio-optimisation problem rather than a deterministic one. Drawing on Mulvaney's Anthropocene framework and IRENA's five pillars, this study quantitatively assesses, using mathematical models, how compounding crises increase systemic risk in India's power sector. The study employs time-series and risk modelling techniques, including AR (1), GARCH (1,1), jump-diffusion, and regime-switching models, to capture persistent volatility, structural breaks, and fat-tailed risks in electricity demand and prices. The analysis suggests that volatility and fat-tailed risk increase the cost of capital, thereby distorting grid investment. Achieving electrification and the 2050 decarbonisation goal requires reductions in variance and tail exposure. Key solutions include regulatory stability, financial de-risking, and improvements to grid infrastructure to address stochastic constraints.

Structure of the Paper and Use of Appendices

This paper integrates empirical evidence with formal modelling. Sections 1–2 introduce the stochastic framework and present both structural and market-based evidence of volatility and tail risk, supported by monthly system indicators (Appendix A) and high-frequency electricity price dynamics (Appendix B). Sections 3–5 develop the portfolio optimisation and energy security framework, with the mathematical formulations provided in the Appendix.

Readers are advised to cross-reference the main text and appendices, where figures, econometric outputs, and model specification are provided to support empirical findings and define stochastic processes. Cross-references are used throughout to direct readers to figures in the appendices, while footnotes are used to guide readers to the corresponding mathematical notations and formulations.

All code and data used for analysis and figure generation are available in the appendices and referenced repositories, ensuring transparency and reproducibility.

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Acronyms and Key Definitions

Acronyms

Acronym	Full Term	Explanation
ETS	Energy Transitions for Sustainability	Framework examining how energy systems transition toward low-carbon and sustainable structures.
IRENA	International Renewable Energy Agency	Intergovernmental organization supporting global renewable energy adoption and policy development.
HHI	Herfindahl–Hirschman Index	A statistical measure of market concentration used to evaluate diversification of energy sources.
AR (1)	Auto-Regressive Model (Order 1)	Time-series model where the current value depends on its immediately previous value.
GARCH	Generalized Auto-Regressive Conditional Heteroskedasticity	Econometric model used to capture volatility clustering in financial or energy price series.
IEX	Indian Energy Exchange	India's largest electricity spot trading platform where electricity prices are discovered through market auctions.
NPV	Net Present Value	The present value of future cash flows discounted by the required rate of return.
CVaR	Conditional Value at Risk	A risk metric that measures expected losses in the worst tail of the distribution beyond a confidence level.
SCC	Social Cost of Carbon	The estimated economic damage caused by emitting one additional unit of CO_2.
PPA	Power Purchase Agreement	Long-term contract between electricity producers and buyers guaranteeing electricity purchase at predetermined prices.
ADB	Asian Development Bank	Multilateral development bank financing infrastructure and energy transition projects in Asia.
MIGA	Multilateral Investment Guarantee Agency	World Bank Group agency providing political risk insurance for infrastructure investments.

Key Definitions

Electrification
The process of expanding electricity access and infrastructure so that households, industries, and services can reliably consume electricity.

Energy Security
The ability of an energy system to ensure a reliable, affordable, and sustainable energy supply while minimising exposure to geopolitical, economic, and environmental risks.

Volatility
The degree of variation in electricity prices or demand over time is, often measured using statistical variance or standard deviation.

Fat Tails
Statistical distributions where extreme events occur more frequently than predicted by a normal distribution. In electricity markets, this refers to rare but severe price spikes.

Stochastic Process
A probabilistic model used to describe systems that evolve over time under uncertainty, such as electricity demand or market prices.

Jump Risk
The possibility that electricity prices experience sudden large movements due to external shocks such as geopolitical events, supply disruptions, or extreme weather.

Energy Portfolio
The combination of different electricity generation technologies (e.g., coal, solar, wind, storage, transmission) used within an energy system.

Stranded Assets
Energy infrastructure that loses economic value prematurely due to regulatory changes, technological shifts, or market transitions.

Grid Flexibility
The ability of the electricity system to adapt to fluctuations in demand and supply, particularly from intermittent renewable energy sources.

1. Brief History of India's Electrification Strategy

India historically focused on expanding generation capacity and connecting households in every state. By 2019, electrification rates had nearly reached full coverage nationwide. The core development goal was achieved. However, apart from the objectives, variables such as quality, reliability, and affordability of electricity have yet to be improved.

1.1 Electrification beyond Deterministic Planning

India's grid structure faces fossil fuel price shocks, transport bottlenecks, geopolitical risks, and environmental constraints due to the reliance on coal (about 70% of generation). With rising demand, policy must anticipate future grid vulnerabilities as the economy grows.

Monthly structural data confirm the dependence of coal and the lack of diversification away from the commodity. The data suggests that coal is the dominant commodity from 2019- 2025, whereas renewable energy (solar, wind, and hydro) rises seasonally but does not displace coal (Appendix A, Figure 1). The Herfindahl-Hirschman Index (HHI) plot (Appendix A, Figure 2), which serves as a quantitative indicator of concentration in the energy mix, supports the trends observed in Figure 1. The data suggest that diversification patterns are seasonal rather than consistently increasing or decreasing. This continues despite the expansion of renewables in India.

Traditional electrification planning models rely on deterministic adequacy conditions, such as ensuring installed capacity exceeds projected peak demand:

$$C \geq D^{\text{peak}}.$$ [1]

This system assumes stable demand distributions and predictable cost structures. Electricity systems usually operate with uncertainty. Demand and prices evolve stochastically:

$$D_t = \mu_D + \sigma_D \, dW_t^D, P_t = \mu_P + \sigma_P \, dW_t^P,$$ [2]

where $W_t$represents a Wiener process and σcaptures volatility. Reliability must therefore be expressed probabilistically:

$$\Pr\left(D_t > C_t\right) \leq \epsilon,$$ [3]

where ϵ denotes acceptable outage probability. Electrification becomes a stochastic control problem rather than a static engineering constraint when distributions shift.

[1] Equation 1: Deterministic Capacity Adequacy Condition
[2] Equation 3: Stochastic Electricity Price Process
[3] Equation 4: Reliability Constraint

With reference to the data, a rolling 12-month volatility of monthly demand growth shows persistent shifts aligned with crisis periods (Appendix A, Figure 3-A4). It can also be observed from (Appendix B, Figure 5]) that electricity prices at a 15min frequency show characteristics of string persistence, volatility clustering, and non-Gaussian tail behaviour. These imply that managing variance and tail risk is imperative to managing electrification quality.

1.2 Mulvaney's Anthropocene Framework

Mulvaney's Anthropocene framework refers to socio-ecological systems. An electricity grid epitomises such a relationship, as it is correlated to global commodity markets, geopolitical shocks, climatic patterns, and fiscal restrictions that produce stochastic outcomes, which justifies why one should model electricity. This reinforces the paper's main argument, which claims that managing volatility and tail risk leads to reliable, affordable, and sustainable electrification during macro crisis events.

2. Compounding Crises and Systematic Risk

2.1 Fossil Fuel and Jump Risk

Electricity spot prices are often modelled as mean-reverting jump-diffusion processes:

$$dS_t = \kappa(\theta - S_t)dt + \sigma S_t\, dW_t + S_t U_t\, dN_t,$$ [4]

In the case of geopolitical risk, both σ and λ increase, thereby raising day-to-day dispersion and the frequency of electricity price spikes.

With reference to the 15 Minute Daily Indian electricity market data in Appendix B, the Auto Regressive [AR (1)] shows a slow decay in shocks. The Generalised Auto Regressive Conditional Heteroskedasticity [GARCH (1,1)] shows strong volatility clustering with close to one (Appendix B, Figure 7]), implying long volatility. The jump-detection model accurately identified rare but extreme price movements (Appendix B, Figure 6]), while the rolling kurtosis showed episodic fat-tailed windows during crisis periods (Appendix B, Figure 8]).

The policy implication is that the main threat to electrification does not come from steady increases in average prices, but from extreme price spikes. These tail events create stress financially and, on the grid, too. Despite being heavily reliant on coal, India is still subject to market shocks from global energy markets through LNG imports and commodity price linkages. This results in external shocks feeding into the domestic energy market.

[4] Equation 5: Electricity Price Jump-Diffusion Model

2.2 Mineral Constraints and Capital Expenditure Variance

Grid expansion depends on metal inputs:

$$CapEx_t = \alpha_1 Cu_t + \alpha_2 Al_t + \alpha_3 Steel_t,$$

with variance increasing in volatility and correlation:

$$\text{Var}\,(CapEx) = \sum_i \alpha_i^2\,\sigma_i^2 + \sum_{i \neq j} \alpha_i\,\alpha_j \rho_{ij} \sigma_i \sigma_j.$$

Post-pandemic disruptions increased both $\sigma_i$and ρ_{ij}. Increasing capital expenditure variance and financing risk, slows down grid investments. From a portfolio perspective, electrification is exposed to operating cash flow volatility (electricity prices) and investment cost volatility (materials), increasing the risk premia for long-term capital investments.

2.3 Pandemic Regime Shifts

Demand distributions can be regime-dependent:

$$D_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma_1^2) & \text{Normal regime} \\ \mathcal{N}(\mu_2, \sigma_2^2) & \text{Crisis regime.} \end{cases}$$ [5]

With reference to (Appendix A, Figure 4), the monthly demand volatility justifies the conclusion of structural shifts. From the IEX data, we can also see clear price regimes that indicate state-dependent risk. This shows to what extent the crisis compounds affect the energy market in India.

2.4 Climate Stress and Negative Covariance

Heatwaves induce:

$$\text{Cov}\,(D_t, C_t) < 0,$$ [6]

Demand (D_t) rises because people use more air conditioning.

Capacity (C_t) effectively falls because thermal plants become less efficient in high temperatures, transmission lines lose carrying capacity, and equipment is stressed.

A negative covariance means these two variables move in opposite directions during stress events, which is dangerous for grid reliability because the system is squeezed from both sides simultaneously. The observed episodes of fat tails in daily price data (Appendix B, Figure 8])

[5] Equation 8: Regime Switching Demand Distribution
[6] Equation 9: Demand-Capacity Covariance

are consistent with these climate-related stress events, while the demand-volatility patterns (Appendix A, Figure 4) help identify the broader periods when the system is under heightened pressure.

3. Energy Security in India Through a Stochastic Risk Lens

This research paper's focus area will also address the five pillars of energy security. Availability, Affordability, Technology Development, Regulatory Factors, and Environmental and Sustainability. This paper will examine these factors through a quantitative lens in a stochastic portfolio optimisation problem.

3.1 Stochastic Portfolio Optimisation

This paper views it as such because the electricity system comprises diverse energy sources and infrastructure choices, including coal, solar, wind, storage, and transmission. Each has different costs, risks, and performance characteristics. It closely resembles the characteristics of a financial portfolio. Hence, the policy problem is not simply to build capacity, but to choose the right mix of technologies to minimise expected system cost while also reducing volatility and downside risk.

3.2 Energy Availability

Energy availability refers to the system's ability to reliably meet demand. Under probabilistic terms, availability requires:

$$\Pr\left(D_t \leq C_t\right) \geq 1 - \epsilon$$

In India, availability is heavily influenced by coal. The monthly data show that coal was the main source of generation from 2019-2024 (Appendix A, Figure 1), with concentration fluctuating seasonally.

From a stochastic perspective, it is about the probability distribution of demand relative to effective capacity under stress. When the demand for energy rises and capacity decreases simultaneously during crisis situations, the probability of scarcity states increases. Hence, availability must be understood as a distributional reliability.

3.3 Affordability

Affordability concerns consumers' ability to access electricity at reasonable, stable prices. Electricity tariffs (the price consumers pay) can be mathematically expressed as:

$$Tariff_t = FuelCost_t + GridCost_t + RiskPremium_t$$ [7]

In practice, political constraints frequently prevent full cost pass-through, leading to:

$$Loss_t = FuelCost_t - Tariff_t$$

From the data, we can deduce that affordability risk is driven less by average price levels than by extreme spikes in electricity prices. This can be seen in (Appendix B, Figure 5]-B4), with extreme price jumps being seen in (Figure 6]). This largely shows that affordability risk is a real problem because of the strain it places on the grid. Policy solutions such as flexible investment and improving market balance can directly reduce volatility.

3.4 Technology Development

This can be defined as the evolution of generation, storage, and grid management technologies. Renewable energy reduces emissions in the long term but introduces intermittency. This means that solar and wind do not produce energy continuously, as they depend on sunlight and wind. This creates variability in supply. Hence, adding more renewables without storage or flexible backup creates supply variability. This necessitates balancing it in the short term.

$$\sigma^2_{\text{system}} = \sigma^2_{\text{renewables}} - \sigma^2_{\text{flex}}$$ [8]

Solutions such as storage, digital forecasting, and transmission expansion reduce conditional variance and reduce fat-tailed volatility. Furthermore, volatility clustering in the IEX daily returns data (Appendix B, Figure 7]) suggests that these solutions should reduce volatility. In addition, volatility shocks can last for days, increasing stress on the grid, increasing financing costs, and discouraging long-term investment. This reinforces the need for such measures in technology development.

[7] Equation 10: Electricity Tariff Structure
[8] Equation 11: System Variance Reduction

3.5 Regulatory Factors

Regulatory factors influence investor confidence, the cost of capital, and the credibility of long-term planning. The required rate of return can be expressed as:

$$r = r_f + \beta \cdot RiskPremium + \gamma_{\text{policy}}$$ [9]

Policy instability increases, raising discount rates and reducing the net present value of infrastructure projects. Political instability refers to the rules governing India's energy market. For instance, sudden changes in renewable tariffs, government intervention in pricing, and uncertain carbon policy. Regulatory uncertainty compounds market risk, and in an environment of high volatility and fat tails, unstable policy further delays investment.

$$OptionValue \propto \sigma^2$$ [10]

Option Value: The benefit of postponing a decision when the future is uncertain.

If volatility in electricity prices is high, investors may delay building a power plant or improving the grid because waiting provides more information to update their prior beliefs about future prices, policy, or demand.

3.5.1 Short-Term Reliability vs Long-Term ETS Goals

With reference to the option values above, delaying renewables and grid improvements increases transition risk. The probability that existing fossil-based assets become economically obsolete can be expressed as:

$$\Pr\,(\text{Stranded}) = f\,(\text{Future Policy,Emissions Trajectory})$$ [11]

If future climate policy tightens or global decarbonisation accelerates, high-emission assets such as coal may become economically unviable.

In India, coal reduces short-term variance (volatility) because of the existing infrastructure supporting it with the supply chain. Yet prolonged reliance on coal delays investments in clean infrastructure, hence increasing long-term volatility and tail events. To summarise, policies that stabilise the system today may cause more systemic risk tomorrow by locking in assets vulnerable to transition shocks.

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[9] Equation 12: Risk-Adjusted Required Return
[10] Equation 13: Real Options Value
[11] Equation 14: Stranded Asset Probability

3.6 Environmental and Social Sustainability

The social cost of electricity generation can be expressed as:

$$\mathbb{E}[C_{\text{social}}] = \mathbb{E}[C_{\text{private}}] + SCC \cdot Emissions$$[12]

It is important to consider externalities as ignoring them leads towards suboptimal portfolio weights and increased long-term systematic risk. Due to the volatility of electricity markets, sustainability remains not simply an environmental goal but also a risk-management one. Decarbonisation reduces long-term transition risk, while climate-resilient infrastructure reduces short-term tail exposure.

4. Finance and Decarbonisation to 2050

Investment decisions follow risk-adjusted discounting:

$$NPV = \sum_{t=1}^{T} \frac{\mathbb{E}[CF_t]}{(1+r)^t},$$[13]

$$r = r_f + \lambda\sigma + \theta\, CVaR_\alpha.$$

Due to the large amounts of volatility and fat tails (Appendix B, Figure 7]-B4), it justifies having the risk premiums added to the equation on both σ and $CVaR_\alpha$. With an overall increase in the rate of return, it reduces the NPV and discourages flexible long-term investment on the grid. Especially when policy risks rise simultaneously.

Reducing systemic variance lowers financing costs:

$$\sigma_{\text{system}} \downarrow \Rightarrow r \downarrow \Rightarrow NPV \uparrow.$$

These can often lead to gaps in investment in energy infrastructure due to mispricing and risk, rather than a lack of capital. This leads to policy solutions using an ETS framework intended to reduce volatility and tail risk.

[12] Equation 15: Social Cost of Electricity
[13] Equation 16: Risk Adjusted Discount Rate

5. ETS Policy Solutions: Reducing Variance and Tail Exposure using ETS Framework

5.1 Regulatory Stability and Market Design

When policies such as renewable procurement rules, tariff structures, or carbon regulations change unpredictably, this causes uncertainty about future cash flows. This results in an increase in risk premiums and cost of capital. Hence, by establishing credible long-term targets, transparent procurement mechanisms such as competitive renewable energy auctions where bids go to the lowest price under long-term contracts and stable tariff frameworks such as automatic fuel cost pass-through adjustments, targeted subsidies for vulnerable consumers instead of broad price caps would stabilise investment expectations.

5.2 Financial De-Risking and Risk Premium Compression

Electricity infrastructure projects require large upfront capital and long payback periods, thereby increasing their risk. High volatility and extreme spikes increase the returns for investors. Solutions include blended finance programmes such as the World Bank's Scaling Solar initiative, political risk guarantees provided by IMGA, concessional loans from institutions such as the ADB, and long-term PPAs used in India's solar auctions through the Solar Energy Corporation of India. Policy-wise, these would reduce revenue volatility and investors' downside risk.

5.3 Infrastructure Modernisation and System Flexibility

Solutions such as grid flexibility, where battery storage, pumped hydro, and improved electricity price forecasting could help balance the volatility out. Transmission expansion reduces supply shortages that can trigger price spikes, while climate-resilient infrastructure reduces the risk of heatwaves affecting the grid in India. From a stochastic perspective, these reduce the variance of system costs and lower the probability of extreme events. A risk-aware planning objective can therefore be expressed as:

$$\min[E(C) + \lambda Var(C) + \theta CVaR_{\alpha}(C)]$$ [14]

An increase in grid flexibility reduces volatility, tail risks, and improves the reliability of the electricity supply.

[14] Equation 17: Risk-Aware Energy Planning Objective

6. Conclusion

India's grid electrification challenge must be viewed as a portfolio optimisation problem with a stochastic nature. Choosing the right assets to fuel electricity generation and avoiding reliance on coal dominance are key to reducing volatility. Monthly structural data show India has persistent coal dominance and oscillating volatility, while daily market data show volatility clustering, price jumps, and macro-event-driven heavy tails. These result in high NPVs, delays in investment, and increased risk for large, long-term investments on the grid.

Electrification policy should therefore minimise expected cost, variance, and tail risk:

$$\min\{\mathbb{E}[C] + \lambda \operatorname{Var}(C) + \theta\, CVaR_{\alpha}(C)\}$$ [15]

In the Anthropocene, electrification is a portfolio allocation problem with correlated, fat-tailed uncertainty. Flexible policy, financial de-risking, and grid improvements would reduce grid volatility and protect it from extreme macro events. These are imperative to achieve a reliable, affordable, and sustainable energy access by 2050.

[15] Equation 17: Risk-Aware Energy Planning Objective

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9. Modelling and Code Repositories

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Reproducibility Statement

All code used for data processing, econometric modelling, and figure generation is publicly available for reproducibility. The repositories include implementations of AR(1), GARCH(1,1), jump detection, rolling higher-moment analysis, and regime-switching models, as well as scripts for constructing structural indicators such as the Herfindahl–Hirschman Index (HHI) and demand volatility measures.

AI Use Disclosure

This manuscript was prepared with limited assistance from generative artificial intelligence tools in accordance with the SMU framework for responsible AI use.

Generative AI tools were used solely for structural and formatting assistance, including the development of the content page, appendix, definitions and acronyms section, and mathematical notation presentation.

No AI tools were used for core analysis, econometric modelling, data processing, or interpretation of results. All modelling, coding, empirical analysis, and policy formulation were conducted independently by the author. The author has critically reviewed all AI-assisted outputs and assumes full responsibility for the accuracy, originality, and integrity of the final submission.

Appendix

Appendix A

Structural Electricity System Indicators (Monthly Data from EMBER)

Appendix A presents structural indicators derived from monthly electricity system data. These figures provide macro-level context for the econometric price dynamics examined in Appendix B. The indicators focus on generation concentration, demand volatility, and stress periods that influence electricity market outcomes.

Figure 1: A1 Structural Transition in Electricity Generation

Data input:
Monthly generation by source (coal, hydro, solar, wind, gas, and others).

Let total generation at time t be:

$$G_t = \sum_{i=1}^{N} G_{i,t}$$

where $G_{i,t}$represents generation from technology i.

The share of each technology is

$$s_{i,t} = \frac{G_{i,t}}{G_t}$$

Interpretation

The figure shows the gradual structural transition in India's electricity mix. Coal remains the dominant generation source, while renewable energy sources, particularly solar and wind, have expanded over time. However, the generation portfolio remains highly concentrated.

From a portfolio perspective, concentration increases systemic exposure to fuel price volatility and regulatory transition risks. Greater diversification of generation technologies would reduce overall system risk by lowering the covariance between energy sources.

Figure 2: A2 Generation Concentration (Herfindahl–Hirschman Index)

Data input:
Technology shares derived from monthly generation data.

Generation concentration is measured using the Herfindahl–Hirschman Index (HHI):

$$HHI_t = \sum_{i=1}^{N} s_{i,t}^2$$

where $s_{i,t}$ is the share of generation technology i.

Interpretation

Higher HHI values indicate greater concentration within the generation mix. The figure shows that the electricity system remains structurally concentrated due to the dominance of coal.

In portfolio terms, concentration increases system risk because shocks affecting a single technology can propagate across the entire electricity system. Diversification through renewables, storage, and flexible generation reduces portfolio variance:

$$Var(C) = \sum_{i} w_i^2 \, \sigma_i^2 + \sum_{i \neq j} w_i \, w_j \sigma_i \sigma_j \rho_{ij}$$

Lower correlations across generation technologies therefore reduce overall system volatility.

Figure 3: 3A Rolling Demand Volatility

Data input:
Monthly electricity demand.

Rolling demand volatility is calculated using a moving-window standard deviation:

$$\sigma_{D,t} = \sqrt{\frac{1}{n-1} \sum_{k=1}^{n} (D_{t-k} - \bar{D})^2}$$

D_t: electricity demand at time t

n: rolling window length

$\bar{D}$: average demand within the window.

Interpretation

The figure shows that electricity demand growth exhibits periods of elevated volatility rather than a smooth trend. These fluctuations reflect macroeconomic cycles, seasonal variation, and weather-driven demand shocks.

Higher demand volatility increases operational stress on the electricity system, particularly when generation capacity and grid flexibility are insufficient to respond quickly to changes in load.

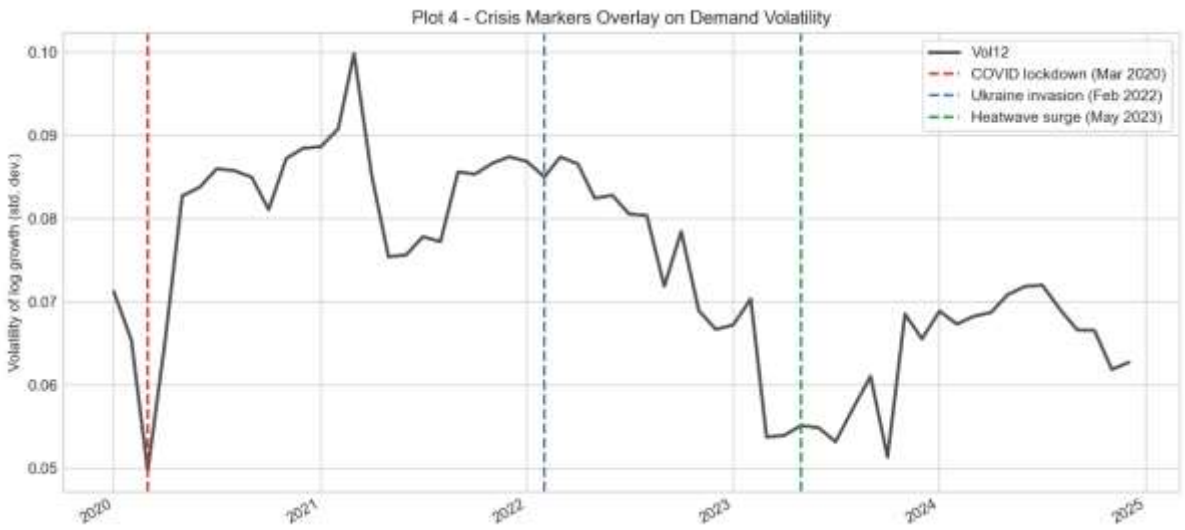

Figure 4: A4 Crisis Period Demand Stress

Data input:
Monthly electricity demand with crisis-period markers (e.g., COVID-19).

Demand dynamics during crises can be described using a regime-switching framework:

$$D_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma_1^2) & \text{Normal regime} \\ \mathcal{N}(\mu_2, \sigma_2^2) & \text{Crisis regime} \end{cases}$$

Interpretation

The figure highlights demand disruptions during crisis periods. These shocks cause sudden drops followed by rapid rebounds in electricity consumption. Such regime shifts increase uncertainty for system planning and contribute to electricity price volatility.

Appendix B

Electricity Price Dynamics (IEX Market Data)

Appendix B analyses electricity price dynamics using daily market clearing prices from the Indian Energy Exchange (IEX). Electricity markets exhibit several stylized statistical features including mean reversion, volatility clustering, jumps, and fat-tailed distributions.

Figure B1: Daily Electricity Price Series

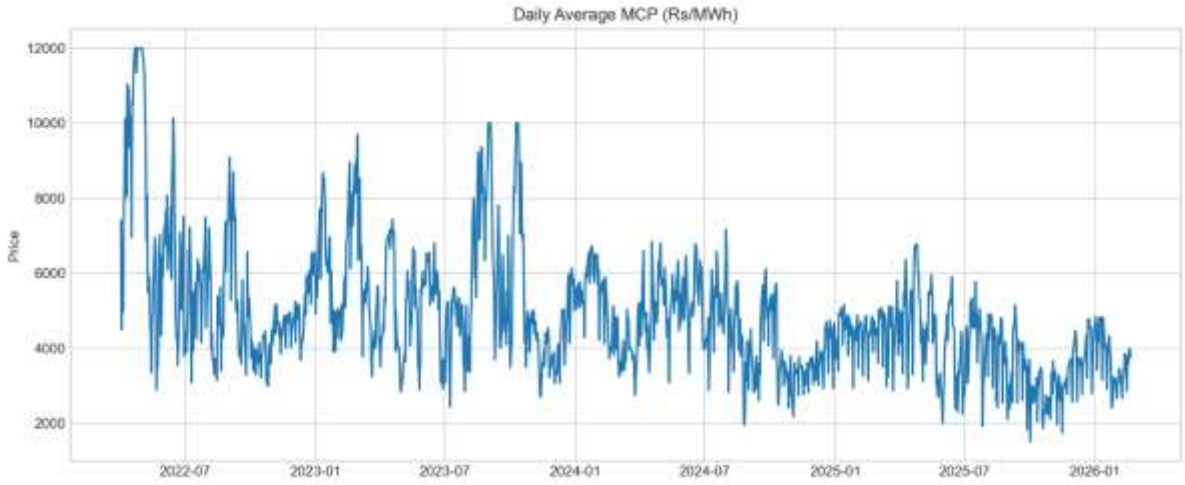

Figure 5: B1 Daily Electricity Price Series

Data input:
Daily market clearing price (MCP) from IEX.

Price dynamics are approximated using an autoregressive model:

$$\log(P_t) = \alpha + \phi \log(P_{t-1}) + \varepsilon_t$$

P_t: electricity price

ϕ: persistence parameter

Interpretation

The price series shows strong persistence and occasional spikes. The AR (1) coefficient indicates that shocks to electricity prices tend to persist over short time horizons.

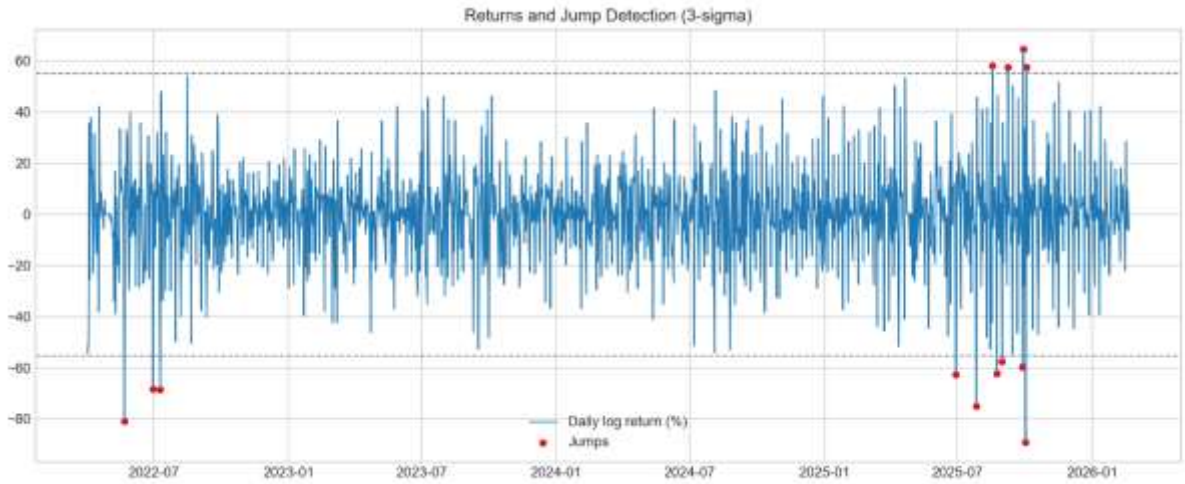

Figure 6: B2 Jump Detection in Price Returns

Returns are defined as

$$r_t = \log(P_t) - \log(P_{t-1})$$

Electricity prices frequently experience sudden spikes caused by supply shortages, demand surges, or transmission constraints.

These dynamics can be modelled using a jump-diffusion process:

$$dS_t = \kappa(\theta - S_t)dt + \sigma dW_t + J_t dN_t$$

where $N_t$represents a Poisson jump process.

Interpretation

The figure shows that electricity returns contain extreme observations consistent with jump behaviour. Such jumps represent discrete supply-demand imbalances rather than continuous market fluctuations.

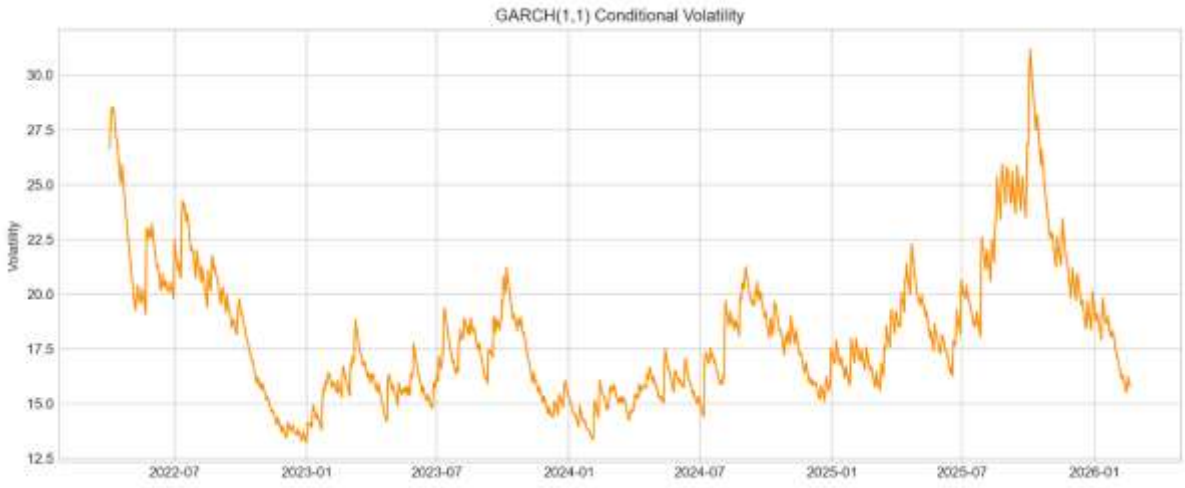

Figure 7: B3 Conditional Volatility (GARCH Model)

Conditional volatility is estimated using a GARCH (1,1) model:

$$r_t = \mu + \varepsilon_t$$
$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

Interpretation

The estimated parameters show strong volatility persistence, with

$$\alpha + \beta \approx 1$$

indicating that volatility shocks decay slowly. Once volatility increases, it tends to remain elevated for extended periods.

This persistence reinforces the need for policies that reduce volatility structurally rather than reacting only to individual price spikes.

Figure 8: B4 Tail Risk and Kurtosis

Tail risk is examined using rolling kurtosis:

$$Kurtosis = \frac{E[(r_t - \mu)^4]}{\sigma^4}$$

Interpretation

Periods of elevated kurtosis indicate fat-tailed distributions, meaning extreme price events occur more frequently than predicted by normal distributions. These tail events are typically associated with system stress, such as supply shortages or extreme weather conditions.

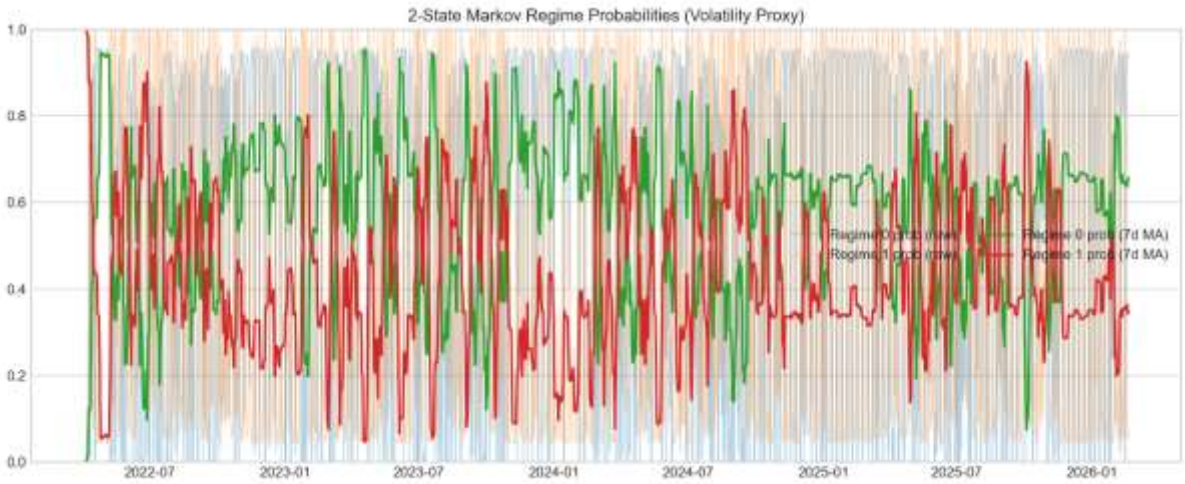

Figure 9: B5 Regime-Switching Price Dynamics

Electricity returns are modelled using a Markov regime-switching framework:

$$r_t \sim \begin{cases} \mathcal{N}(\mu_0, \sigma_0^2) & \text{Low-volatility regime} \\ \mathcal{N}(\mu_1, \sigma_1^2) & \text{High-volatility regime} \end{cases}$$

with transition probabilities

$$P_{ij} = \Pr\left(S_t = j \mid S_{t-1} = i\right)$$

Interpretation

The system spends most of its time in the low-volatility regime but occasionally transitions into a high-volatility regime characterized by extreme price spikes and increased uncertainty.

Appendix A & B Conclusion

The structural indicators in Appendix A and the econometric results in Appendix B demonstrate that electricity markets exhibit persistent volatility, episodic tail events, and regime-switching dynamics. These empirical findings support the paper's central argument that electrification strategies must incorporate risk management and volatility reduction rather than relying solely on deterministic capacity expansion.

Appendix C

Mathematical Notation and Formula Explanations

Appendix C explains the mathematical models and notation used throughout the report. The formulas are presented in the same order as they appear in the main document and are indexed according to the corresponding chapter and subsection numbers.

C1. Section 1.1 – Deterministic Capacity Adequacy Condition

Traditional electrification planning assumes that installed generation capacity must exceed peak electricity demand.

$$C \geq D^{peak}$$

Variables

Symbol	Definition
C:	Installed electricity generation capacity
D^{peak}:	Maximum expected electricity demand

Explanation

This deterministic planning rule ensures that electricity supply exceeds expected peak demand. However, it assumes demand is predictable and stable. Electricity systems operate under uncertainty, requiring probabilistic reliability measures.

Equation 1: Deterministic Capacity Adequacy Condition

C2. Section 1.1 – Stochastic Demand Process

Electricity demand evolves under uncertainty and can be represented using a stochastic process.

$$D_t = \mu_D + \sigma_D dW_t^D$$

Variables

Symbol	Definition
D_t:	Electricity demand at time t
μ_D:	Mean demand level
σ_D:	Demand volatility
W_t^D:	Wiener process representing random shocks

Explanation

This model captures the random fluctuations in electricity demand caused by weather conditions, economic activity, and other systemic shocks.

Equation 2: Stochastic Demand Process

C3. Section 1.1 – Stochastic Electricity Price Process

Electricity prices also evolve stochastically.

$$P_t = \mu_P + \sigma_P dW_t^P$$

Variables

Symbol	Definition
P_t:	Electricity price at time t
μ_P:	Mean price level
σ_P:	Price volatility
W_t^P:	Wiener process representing random shocks

Explanation

This formulation reflects the uncertain evolution of electricity prices driven by supply disruptions, fuel costs, and demand fluctuations.

Equation 3: Stochastic Electricity Price Process

C4. Section 1.1 – Reliability Constraint

Electricity system reliability can be expressed probabilistically.

$$Pr(D_t > C_t) \leq \epsilon$$

Variables

Symbol	Definition
D_t:	Electricity demand
C_t:	Available generation capacity
ϵ:	Acceptable probability of supply shortage

Explanation

Rather than guaranteeing supply deterministically, planners allow a small probability of outages.

Equation 4: Reliability Constraint

C5. Section 2.1 – Electricity Price Jump-Diffusion Model

Electricity prices often exhibit mean reversion combined with sudden spikes.

$$dS_t = \kappa(\theta - S_t)dt + \sigma S_t dW_t + S_t U_t dN_t$$

Variables

Symbol	Definition
S_t:	Electricity spot price
κ:	Speed of mean reversion
θ:	Long-run equilibrium price
σ:	Diffusion volatility
W_t:	Brownian motion
N_t:	Poisson jump process
U_t:	Jump magnitude

Explanation

This model captures three important characteristics of electricity markets:

- Mean-reverting prices
- Continuous volatility
- Sudden price spikes caused by supply shocks or extreme demand.

Equation 5: Electricity Price Jump-Diffusion Model

C6. Section 2.2 – Grid Capital Expenditure Model

Grid investment costs depend on commodity inputs used in infrastructure construction.

$$CapEx_t = \alpha_1 Cu_t + \alpha_2 Al_t + \alpha_3 Steel_t$$

Variables

Symbol	Definition
Cu_t:	Copper price
Al_t:	Aluminium price
$Steel_t$:	Steel price
α_i:	Commodity input weights

Explanation

Transmission infrastructure requires large quantities of copper, aluminium, and steel. Variations in commodity prices affect total infrastructure investment costs.

Equation 6: Grid Capital Expenditure Model

C7. Section 2.2 – Capital Expenditure Variance

The uncertainty in capital costs depends on commodity price volatility and correlations.

$$Var(CapEx) = \sum_{i} \alpha_i^2 \, \sigma_i^2 + \sum_{i \neq j} \alpha_i \, \alpha_j \rho_{ij} \sigma_i \sigma_j$$

Variables

Symbol	Definition
σ_i:	Commodity price volatility
ρ_{ij}:	Correlation between commodities
α_i:	Input weights

Explanation

Higher volatility and stronger correlations between commodity prices increase uncertainty in grid investment costs.

Equation 7: Capital Expenditure Variance

C8. Section 2.3 – Regime Switching Demand Distribution

Electricity demand may follow different statistical regimes.

$$D_t \sim \begin{cases} \mathcal{N}(\mu_1, \sigma_1^2) & \text{Normal regime} \\ \mathcal{N}(\mu_2, \sigma_2^2) & \text{Crisis regime} \end{cases}$$

Variables

Symbol	Definition
μ_1, σ_1:	Mean and volatility in normal conditions
μ_2, σ_2:	Mean and volatility during crisis conditions

Explanation

Demand volatility can increase significantly during crises such as pandemics, geopolitical shocks, or extreme weather events.

Equation 8: Regime Switching Demand Distribution

C9. Section 2.4 – Demand-Capacity Covariance

Heatwaves can produce a negative relationship between demand and available capacity.

$$Cov(D_t, C_t) < 0$$

Variables

Symbol	Definition
D_t:	Electricity demand
C_t:	Available generation capacity

Explanation

During heatwaves:

- Electricity demand increases due to air conditioning.
- Thermal plant efficiency decreases and transmission capacity falls.
- This creates simultaneous stress on the electricity system.

Equation 9: Demand-Capacity Covariance

C10. Section 3.3 – Electricity Tariff Structure

Electricity tariffs can be decomposed into cost components.

$$Tariff_t = FuelCost_t + GridCost_t + RiskPremium_t$$

Variables

Symbol	Definition
$FuelCost_t$:	Cost of fuel used for electricity generation
$GridCost_t$:	Cost of transmission and distribution
$RiskPremium_t$:	Compensation for volatility and financial risk

Explanation

Electricity tariffs reflect the underlying costs of fuel, infrastructure, and financial risk.

Equation 10: Electricity Tariff Structure

C11. Section 3.4 – System Variance Reduction

System volatility depends on renewable variability and flexibility technologies.

$$\sigma_{system}^2 = \sigma_{renewables}^2 - \sigma_{flex}^2$$

Variables

Symbol	Definition
$\sigma_{renewables}$:	Variability of renewable generation
σ_{flex}:	Stabilizing effect of storage and flexibility technologies

Explanation

Storage, flexible generation, and transmission expansion reduce system volatility.

Equation 11: System Variance Reduction

C12. Section 3.5 – Risk-Adjusted Required Return

Investment returns incorporate market and policy risk.

$$r = r_f + \beta \cdot RiskPremium + \gamma_{policy}$$

Variables

Symbol	Definition
r:	Required rate of return
r_f:	Risk-free rate
β:	Market risk exposure
γ_{policy}:	Regulatory risk premium

Explanation

Policy uncertainty increases financing costs for infrastructure investments.

Equation 12: Risk-Adjusted Required Return

C13. Section 3.5 – Real Options Value

Investment delay becomes more valuable when volatility increases.

$$OptionValue \propto \sigma^2$$

Variables

Symbol	Definition
$OptionValue$:	Value of delaying investment
σ:	Market volatility

Explanation

Higher volatility increases the value of waiting before committing capital to large infrastructure projects.

Equation 13: Real Options Value

C14. Section 3.5.1 – Stranded Asset Probability

The probability that assets become economically obsolete depends on future policy and emissions trajectories.

$$Pr(Stranded) = f(FuturePolicy, EmissionsTrajectory)$$

Explanation

If climate policies tighten or decarbonisation accelerates, fossil-based infrastructure may lose economic viability.

Equation 14: Stranded Asset Probability

C15. Section 3.6 – Social Cost of Electricity

Total social cost includes private production costs and environmental externalities.

$$E[C_{social}] = E[C_{private}] + SCC \cdot Emissions$$

Variables

Symbol	Definition
$C_{private}$:	Direct electricity generation cost
SCC:	Social cost of carbon
$Emissions$:	Carbon emissions

Explanation

Ignoring environmental externalities leads to suboptimal energy portfolio choices.

Equation 15: Social Cost of Electricity

C16. Section 4 – Risk Adjusted Discount Rate

Energy infrastructure investments incorporate volatility and tail risk.

$$NPV = \sum_{t=1}^{T} \frac{\mathbb{E}[CF_t]}{(1+r)^t},$$

$$r = r_f + \lambda\sigma + \theta CVaR_\alpha$$

Variables

Symbol	Definition
r:	Required rate of return
r_f:	Risk-free rate
σ:	Volatility
$CVaR_\alpha$:	Tail risk
λ, θ:	Risk aversion parameters

Explanation

This formulation adjusts the discount rate to explicitly price uncertainty, where higher volatility and tail risk increase the required return on energy investments. As a result, projects exposed to unstable electricity markets face a higher cost of capital, reducing their net present value and discouraging investment.

Equation 16: Risk Adjusted Discount Rate

C17. Section 5 – Risk-Aware Energy Planning Objective

Energy planning can be expressed as a stochastic optimisation problem.

$$\min[E(C) + \lambda Var(C) + \theta CVaR_{\alpha}(C)]$$

Variables

Symbol	Definition
$E(C)$:	Expected system cost
$Var(C)$:	Variance of system cost
$CVaR_{\alpha}$:	Tail risk
λ, θ:	Risk weights

Explanation

The objective is to minimise expected costs while penalising volatility and extreme downside risk.

Equation 17: Risk-Aware Energy Planning Objective